

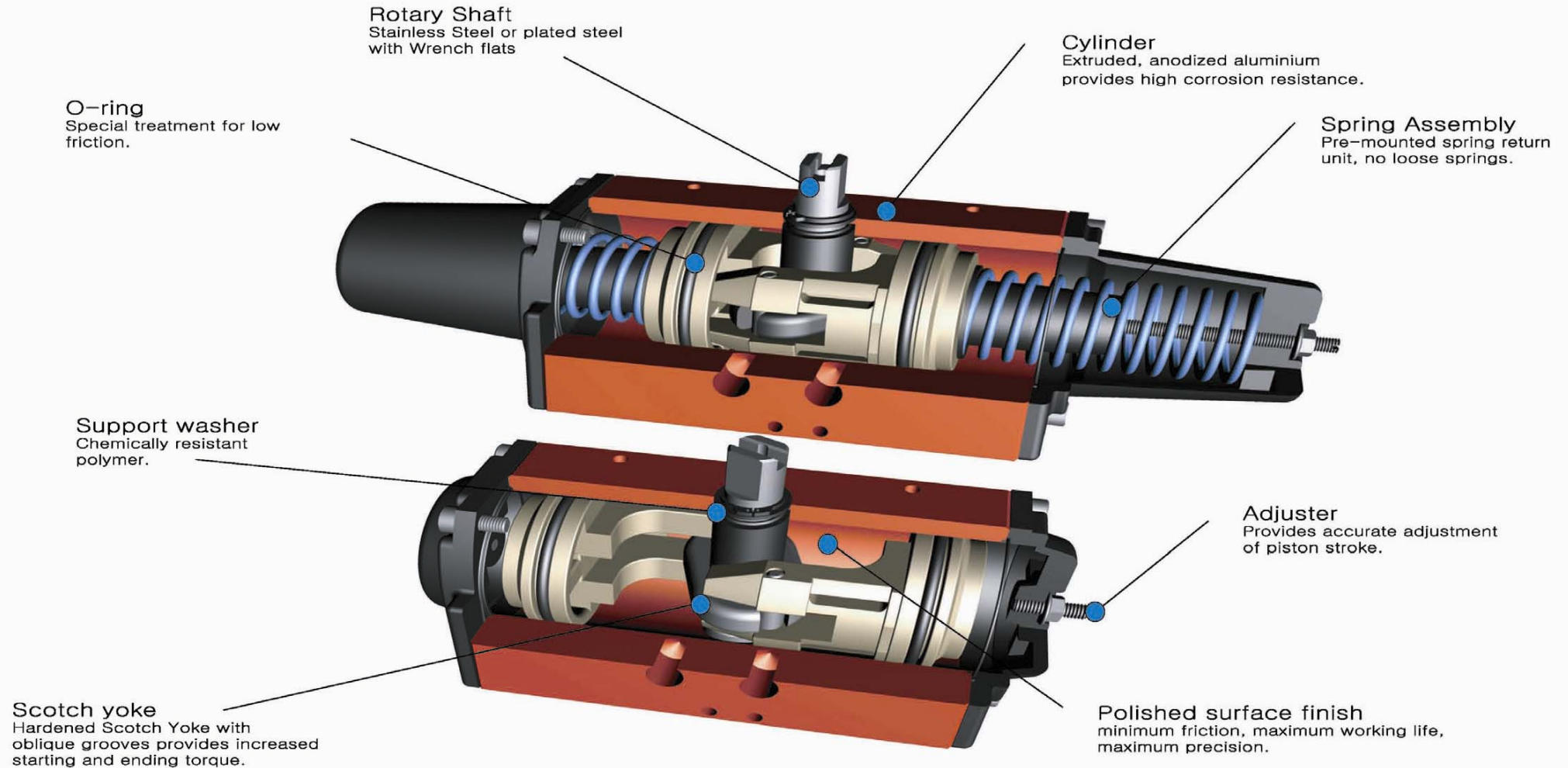
Pneu - Max[®]



ISO 9001 : 2000

**Pneumatic Actuator SCOTCH-YOKE Type
PM-S Series**

The *Pneu-Max*[®] Actuator of PM-S Series



The *Pneu-Max*[®] rotary actuator are designed to operate rotating valve.

The entire series features robust design / construction, light weight and compact overall dimension in proportion to the torque delivered.

Butterfly, Ball, Plug & Segment ball valve for control as well as on-off service. PM-S050~S160 is scotch-yoke mechanism which produces a torque curve that most closely match as the torque requirement of rotating valves.

Low-friction seals ensure smooth, rotary motion and require no additional lubrication, thereby reducing maintenance cost.

Actuators in the PM-S series are double acting operated by air in both directions or single acting with spring return in on direction.

Both single and double acting models are available in a wide range of size and design.

PM-S series actuators are permanently lubricated with specially selected environmental friendly lubricants.

General Features & Connection

PM-S Series General Features

- # Scotch Yoke Double cylinder & Double piston
 - Ideal high torque
- # Low friction materials are aluminum
 - High response
- # Main materials are aluminum(cylinder, cover)
 - Compact and Low Weight
- # Easy-Changeable & Maintenance
 - Good cangeability between spring return cylinder and double acting cylinder at field
- # ISO Standard
 - Actuator mounting pad are provided under ISO 5211 standard
- # NAMUR standard/VDI/VDE 3845
 - NAMUR standard connection for direct mounting of solenoid valve, limits s/w (box type), positioner
- # Supply – Dry or lubricated filtered compressed air
- # Cylinder inside surface finished – To minimize friction and to maximize the life of the actuator
- # The drive shaft key – According to ISO 5211/DIN 3337 standards
- # Ambient temperatures – $-20^{\circ}\text{C} \sim 80^{\circ}\text{C}$ / # Max pressure – 10barG



Pneu-Max has been manufacturing Pneumatic actuators in Korea since 2000.

PM-S Series Connection



Air supply connection is designed in accordance with NAMUR standard to install solenoid valves.



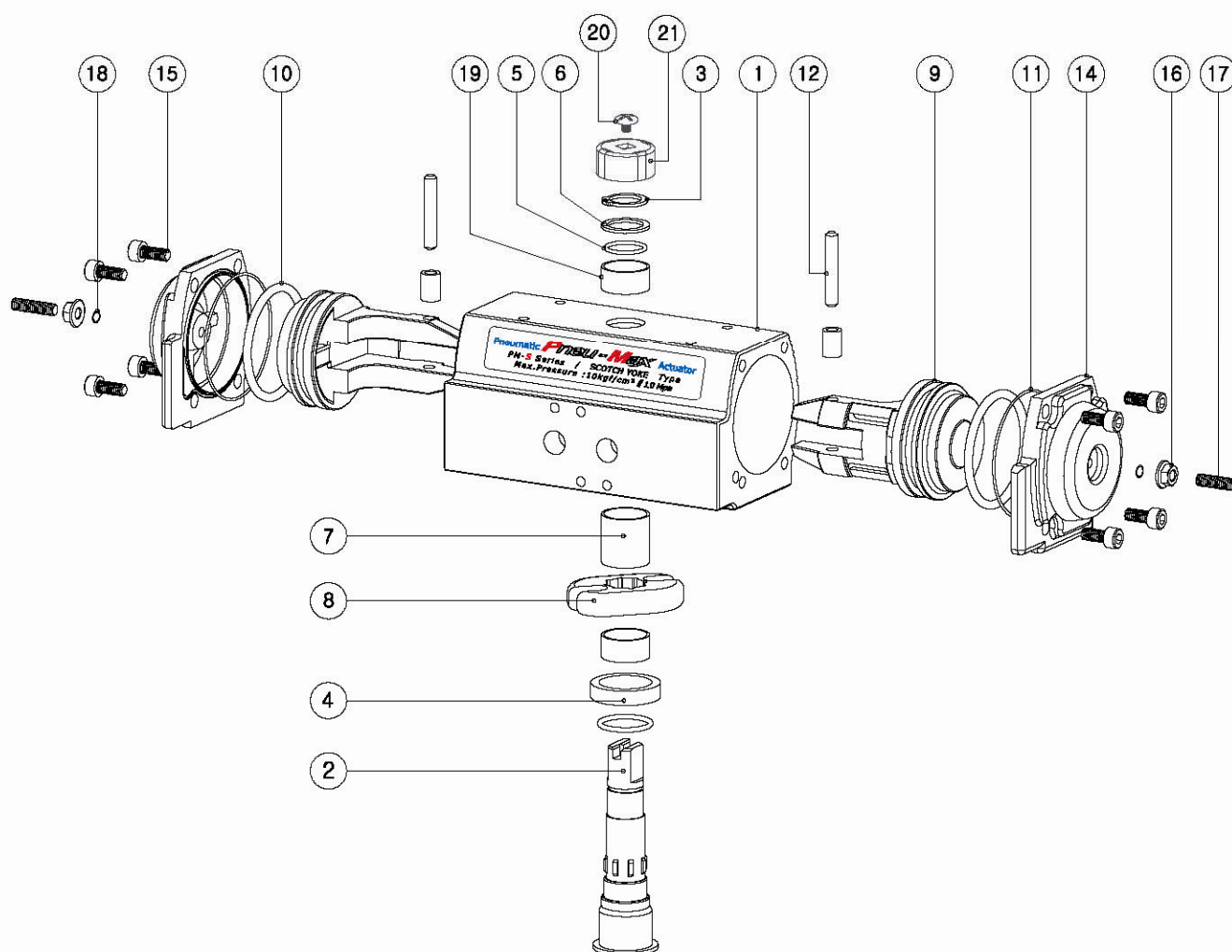
The NAMUR drive pinion and NAMUR top mounting connection permit direct installation of accessories such as limits switch box and positioner.



Bottom mounting connection is designed in accordance with ISO 5211 and DIN 3337 standards for direct mounting with valve gear box or brackets.

Assembly Drawing & Part List

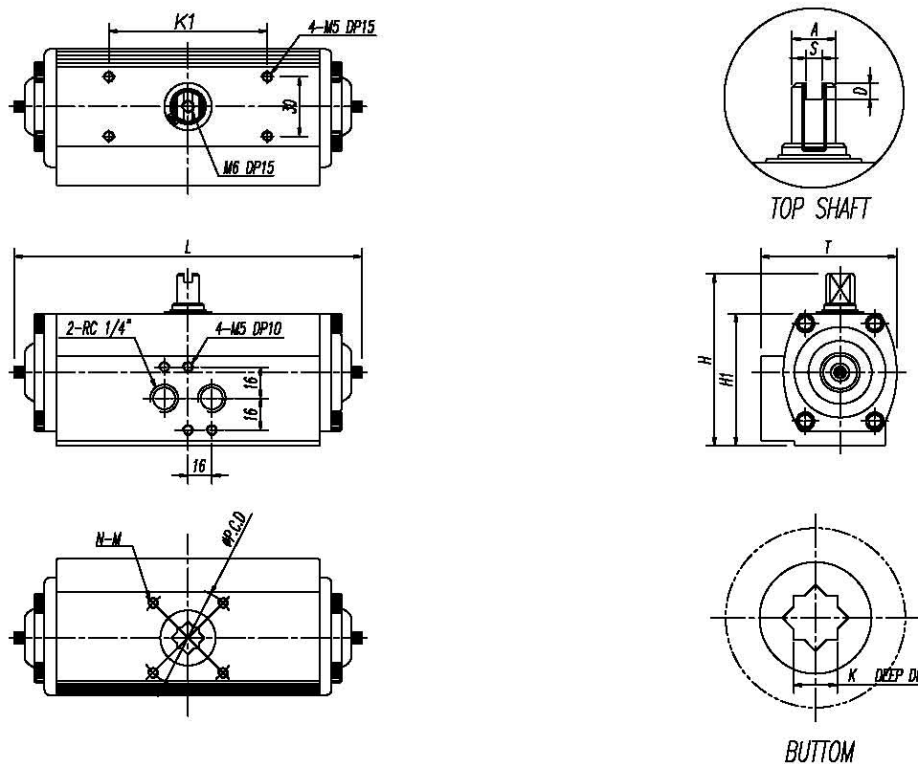
PM-S Series Double Acting Actuator Part List & Table



NO.	DESCRIPTION	Q'ty	STANDARD MATERIAL	REMARK
1	Body (Cylinder)	1	Alluminum	Anodized
2	Shaft	1	Stainless Steel	
3	Snap Ring	1	Stainless Steel	
4	Bearing Bushing	1	Polymer Material	
5	O-ring (Shaft Top)	1	NBR	
6	Washer (Outside)	1	Stainless Steel	
7	Guide (shaft Bottom)	1	Polymer Material	
8	Crank	1	Steel	
9	Piston	2	Engineering Plastics	
10	O-ring (Piston)	2	NBR	
11	O-ring (End Cap)	2	NBR	
12	Pin	2	Steel	
13	Pin Bushing	2	Steel	
14	End Cap	2	Alluminum	Anodized
15	Cap Bolt	8	Stainless Steel	
16	Nut (Adjust Screw)	2	Stainless Steel	
17	Adjust Screw	2	Steel	
18	O-ring (Adjust Screw)	2	NBR	
19	Guide (shaft Top)	1	Polymer Material	
20	Indicator Screw	1	Steel	
21	Indicator	1	Plastic	

Dimension & Torque Table

PM-S Series Double Acting Dimension Table



(Unit : mm)

MODEL	L	T	A	S	D	N-M	P.C.D	H1	H	K1	K	DP	Weight(kg)
050D	188	69	11	4	4	M5/M6	50	67	87	80	11 × 11	13	1.5
065D	234	83	13	4	4	M6/M8	50/70	86	108	80	14 × 14	17	2.5
080D	286	98	17	4	4	M8	70	103	123	80	17 × 17	19	4
100D	344	114	22	4	4	M8/M10	70/102	123	143	80	22 × 22	26	7
125D	443	136	22	4	4	M8/M10	70/102	144	164	80	22 × 22	26	11
140D	486	158	22	4	4	M10/M12	102/125	160	180	80	27 × 27	30	18
160D	560	178	26	4	4	M10/M12	102/125	190	210	80	27 × 27	30	25
						M16	140				36 × 36		
210D	620	230	36	4	4	M20	165	261	290	130	36 × 36	50	44.5
											46 × 46	60	

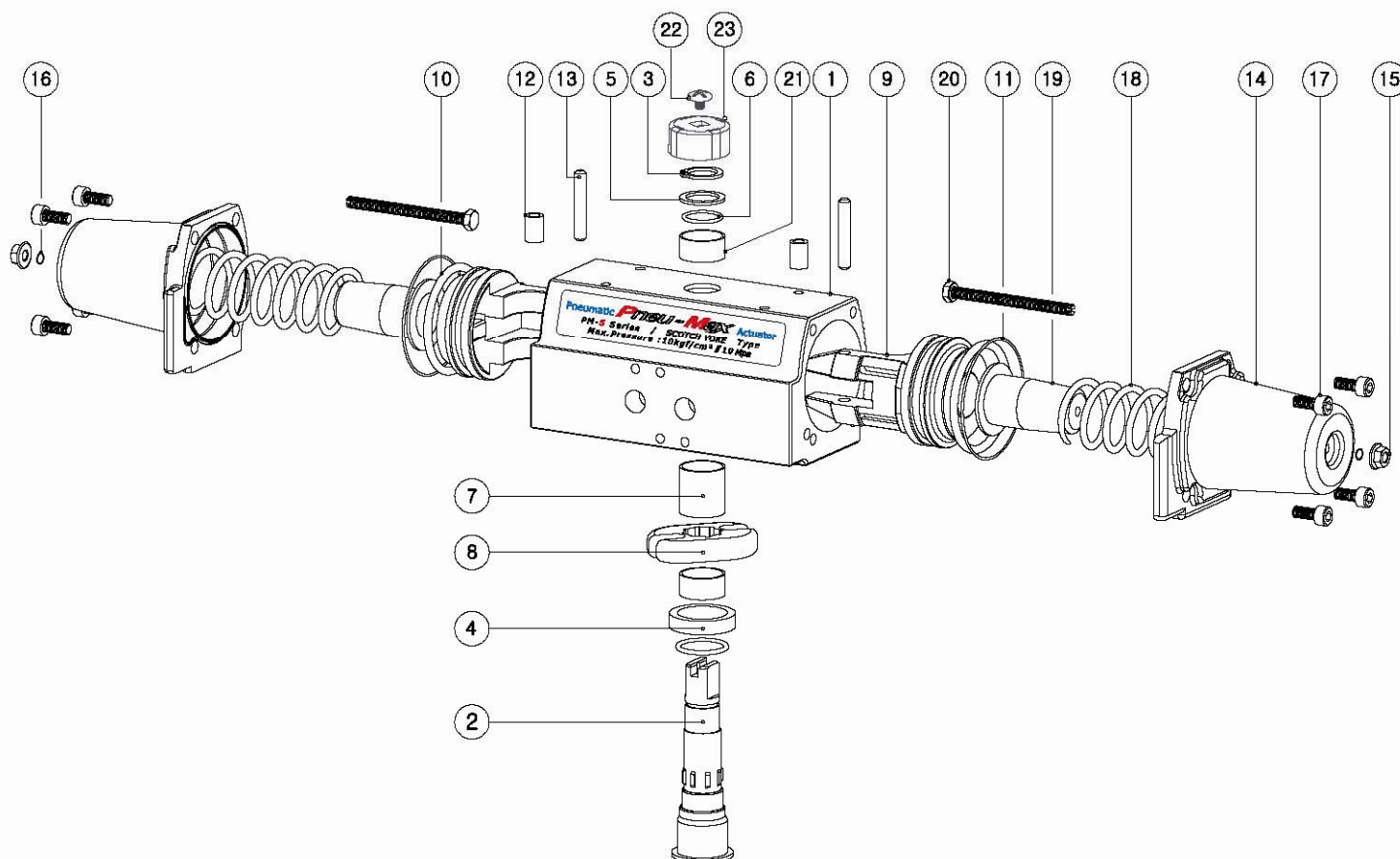
PM-S Series Double Acting Torque Table

Unit : kgf-m

MODEL	DEGREE	SUPPLY PRESS. AIR : 4 kgf/cm ² G		SUPPLY PRESS. AIR : 4.5 kgf/cm ² G		SUPPLY PRESS. AIR : 5.0 kgf/cm ² G		SUPPLY PRESS. AIR : 6.0 kgf/cm ² G	
		AIR to CLOSE	AIR to OPEN	AIR to CLOSE	AIR to OPEN	AIR to CLOSE	AIR to OPEN	AIR to CLOSE	AIR to OPEN
050D	0°	3.7	2.8	3.8	3.3	3.8	3.7	4.2	4.1
	45°	2.0	2.1	2.3	2.4	2.5	2.6	3.0	3.1
	90°	3.3	3.8	3.6	3.9	3.9	4.0	5.0	5.2
065D	0°	8.5	7.8	9.5	8.7	10.5	9.5	13.1	11.6
	45°	4.5	4.9	5.1	5.6	5.7	6.3	6.8	7.3
	90°	7.2	7.1	8.0	8.2	8.8	9.3	10.7	10.7
080D	0°	16.4	14.3	16.0	16.3	15.6	18.3	26.1	21.0
	45°	8.4	9.2	9.8	10.7	11.2	12.1	13.0	14.4
	90°	13.0	13.3	15.1	14.5	17.1	15.6	19.8	20.9
100D	0°	26.5	24.7	29.9	27.0	33.2	29.2	39.3	36.8
	45°	14.7	16.5	16.6	18.1	18.5	19.6	23.7	25.0
	90°	23.1	23.7	26.4	26.9	29.7	30.1	34.8	36.3
125D	0°	57.6	55.1	65.1	61.5	72.5	67.8	88.7	79.4
	45°	33.9	36.0	38.2	40.2	42.5	44.3	51.3	53.1
	90°	51.0	49.1	57.3	55.2	63.5	61.3	75.0	73.2
140D	0°	83.0	75.4	96.8	83.9	110.5	92.3	131.1	109.7
	45°	45.0	45.5	51.5	51.9	57.9	58.2	69.1	66.8
	90°	68.9	72.0	77.6	80.0	86.3	88.0	105.5	97.0
160D	0°	114.7	106.2	131.6	118.7	148.5	131.2	180.0	160.0
	45°	73.0	75.8	82.5	85.4	92.0	94.9	111.0	113.3
	90°	117.0	132.0	130.9	147.8	144.8	163.5	175.0	202.7
210D	0°	280.0	260.0	294.0	286.0	330.0	320.0	392.0	382.0
	45°	146.0	150.0	171.0	175.0	192.0	195.0	230.0	240.0
	90°	215.0	230.0	260.0	271.0	280.0	290.0	335.0	345.0

Assembly Drawing & Part List

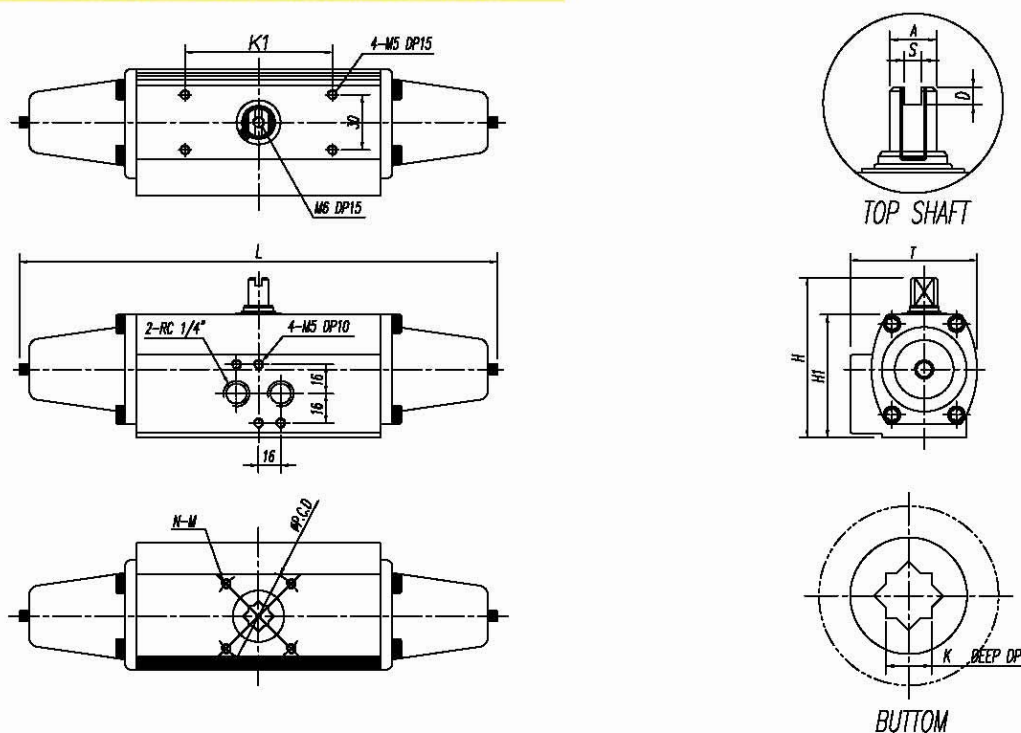
PM-S Series Spring Return Actuator Part List & Table



NO.	DESCRIPTION	Q'ty	STANDARD MATERIAL	REMARK
1	Body (Cylinder)	1	Alluminum	Anodized
2	Shaft	1	Steel	
3	Snap Ring	1	Stainless Steel	
4	Bearing Bushing	1	Polymer Material	
5	Washer (Outside)	1	Stainless Steel	
6	O-ring (Shaft Top)	1	NBR	
7	Guide (shaft Bottom)	1	Polymer Material	
8	Crank	1	Steel	
9	Piston	2	Engineering Plastics	
10	O-ring (Piston)	2	NBR	
11	O-ring (End Cap)	2	NBR	
12	Pin Bushing	2	Steel	
13	Pin	2	Steel	
14	End Cap	2	Alluminum	Anodized
15	Nut (Adjust Screw)	2	Stainless Steel	
16	O-ring (Adjust Screw)	2	NBR	
17	Cap Bolt	8	Stainless Steel	
18	Spring	2	Spring Steel	Corrosion Protected
19	Spring Seat	2	Steel	
20	Adjust Screw	2	Steel	
21	Guide (shaft Top)	1	Polymer Material	
22	Indicator Screw	1	Stainless Steel	
23	Indicator	1	Plastic	

Dimension & Torque Table

PM-S Series Spring Return Dimension Table



(Unit : mm)

MODEL	L	T	A	S	D	N-M	P.C.D	H1	H	K1	K	DP	Weight(kg)
050D	258	69	11	4	4	M5/M6	50	67	87	80	11 × 11	13	1.6
065D	320	83	13	4	4	M6/M8	50/70	86	108	80	14 × 14	17	3.2
080D	418	98	17	4	4	M8	70	103	123	80	17 × 17	22	5.4
100D	506	114	22	4	4	M8/M10	70/102	123	143	80	22 × 22	26	10
125D	649	136	22	4	4	M8/M10	70/102	144	164	80	22 × 22	26	17
140D	716	158	22	4	4	M10/M12	102/125	160	180	80	27 × 27	30	24
160D	850	178	26	4	4	M10/M12	102/125	190	210	80	27 × 27	30	35
						M16	140				36 × 36		
210D	980	230	36	4	4	M20	165	261	290	130	36 × 36	50	72.5
											46 × 46	60	

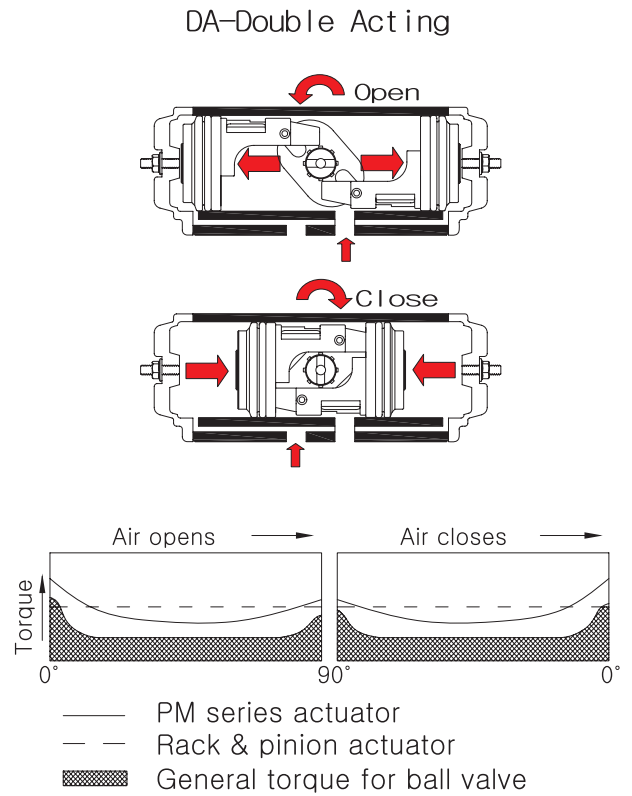
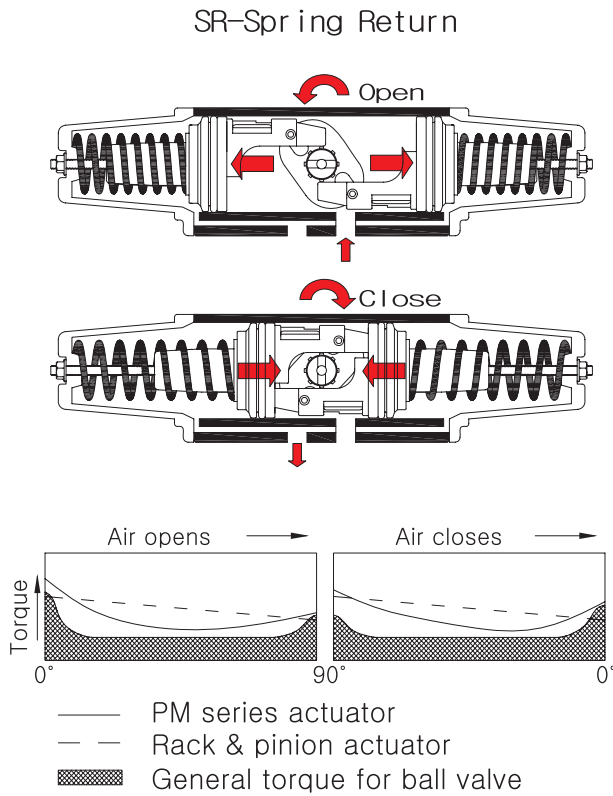
PM-S Series Spring Return Torque Table

Unit : kgf-m

MODEL	DEGREE	SPRING TORQUE (1~2 kgf/cm2G)		SPRING TORQUE (2~3 kgf/cm2G)		SPRING TORQUE (3~4 kgf/cm2G)	
		SPRING to CLOSE	AIR to OPEN (3kgf/cm2G)	SPRING to CLOSE	AIR to OPEN (4.5kgf/cm2G)	SPRING to CLOSE	AIR to OPEN (6kgf/cm2G)
050S	0°	1.0	2.0	1.4	2.6	1.7	3.4
	45°	0.9	0.9	1.2	1.2	1.5	1.5
	90°	2.0	1.0	2.6	1.4	3.4	1.7
065S	0°	2.2	3.9	3.5	5.4	4.8	7.0
	45°	1.8	1.8	2.3	2.8	3.7	3.2
	90°	3.9	2.2	5.4	3.5	7.0	4.8
080S	0°	4.0	7.0	6.0	10.0	8.0	13.0
	45°	3.0	3.0	5.0	5.0	7.0	7.0
	90°	7.0	4.0	10.0	6.0	13.0	8.0
100S	0°	7.0	14.0	10.0	19.0	13.0	24.0
	45°	5.0	5.0	8.0	8.0	11.0	11.0
	90°	14.0	7.0	19.0	10.0	24.0	13.0
125S	0°	15.0	25.0	23.0	38.0	30.0	51.0
	45°	13.0	13.0	19.0	19.0	26.0	26.0
	90°	25.0	15.0	38.0	23.0	51.0	30.0
140S	0°	20.0	37.0	30.0	55.0	41.0	73.0
	45°	17.0	17.0	29.0	29.0	34.0	34.0
	90°	37.0	20.0	55.0	30.0	73.0	41.0
160S	0°	40.0	54.0	55.0	75.0	77.0	115.0
	45°	29.0	29.0	42.0	42.0	56.0	56.0
	90°	54.0	40.0	75.0	55.0	115.0	77.0
210S	0°	67	120.0	105.0	162.0	142.0	217.0
	45°	56	56.0	94.0	94.0	126.0	126.0
	90°	120	67.0	162.0	105.0	217.0	142.0

Cylinder Operating & Air Consumption

PM-S Series Spring Return & Double Acting Operating



PM-S Series Calculation for Air Consumption

< Spring Return Actuator >

$$V = B \times \left(\frac{P+1}{1} \right) \times N$$

< Double acting Actuator >

$$V = (B + A) \times \left(\frac{P+1}{1} \right) \times N$$

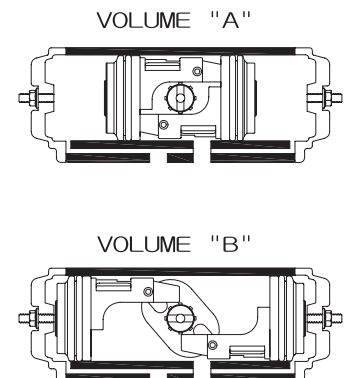
V = Air Consumption
(N Liter)

P = Supply Air Pressure
(kgf/cm²)

A,B = Cylinder Volume
(Liter)

N = Operation Time

MODEL		CYLINDER VOLUME(L)	
		A	B
050S	050D	0.13	0.11
065S	065D	0.25	0.24
080S	080D	0.54	0.48
100S	100D	0.97	0.86
125S	125D	1.78	1.68
140S	140D	2.74	2.45
160S	160D	4.22	3.74
210S	210D	9.97	6.19



Manual override unit

PM-S Series Side Handwheel – "S" Type



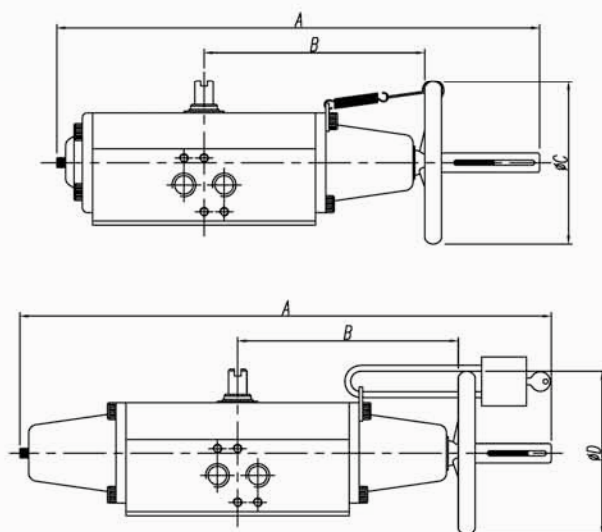
For double acting and spring return actuator, sizes 050~160.

Leaves the top of the actuator free for the fitting of accessories.

Enables locking of the actuator in open & close position. Especially useful with spring return units spring return.

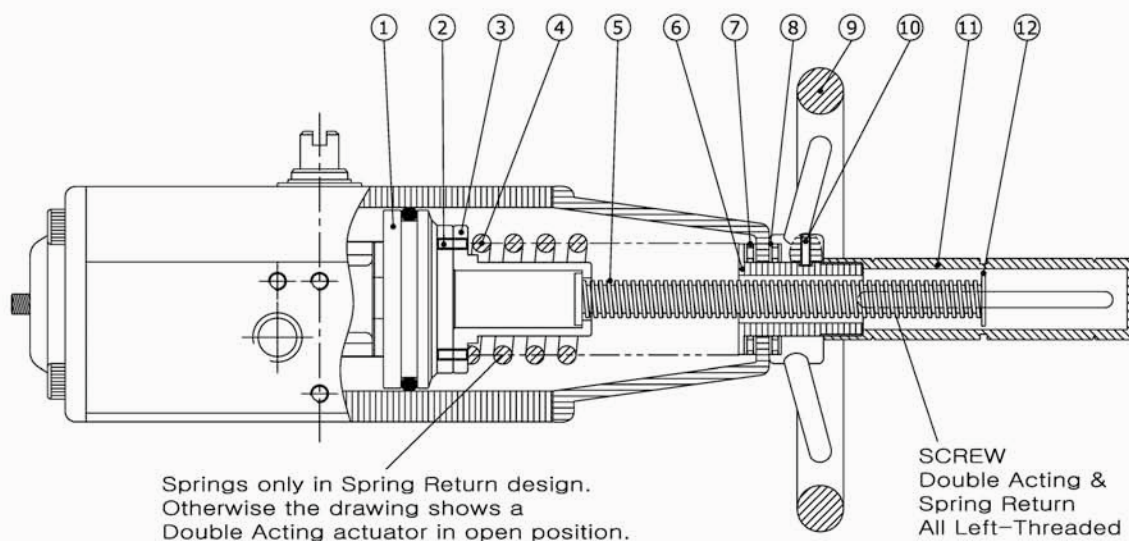
The handwheel is standing still during Pneu-Max operation.

Simple and secure function.



MODEL	Double Acting "A"	Spring Return "A"	B	C	D
050	198	310	150	110	110
065	254	395	180	160	160
080	282	508	230	200	200
100	382	581	285	250	250
125	478	732	355	250	250
140	515	811	390	400	400
160	970	964	450	400	415

PM-S Series Side Handwheel Part List & Table



NO.	PART	NO.	PART	NO.	PART
1	Piston	5	Screw Stem	9	Hand Wheel
2	Screw Bolt	6	Stem Nuts	10	Set Screw
3	Spring Guide	7	Needle Roller Bearing	11	Protecting Tube
4	Spring	8	Plate	12	Indicator

Pneu-Max Modeling & Product Pictures

Pneu-Max series Modeling

PM-S SERIES 050 S 10-S

ACTUATOR MODEL

R series	Rack & Pinion
S series	Scotch Yoke
H series	Heavy Duty
Q series	Rotary Motor Actuator
L series	Linear Motor Actuator

ACTUATOR SIZE

050
065
080
100
125
140
160
210

BODY MODEL

10	Ball (Float) Valve
11	Ball (Trunnion) Valve
12	1-Piece Ball Valve
13	3-Piece Ball Valve
14	3-Way Ball Valve
15	Y-Type Ball Valve
21	Butterfly Valve
22	High-Performance Butterfly Valve
23	3-Way Butterfly Valve
24	Damper Valve
33	V-Notch Valve
40	Globe Valve
50	Gate Valve
51	Knife Gate Valve
52	Plug Valve
60	Diaphragm Valve

OPTION

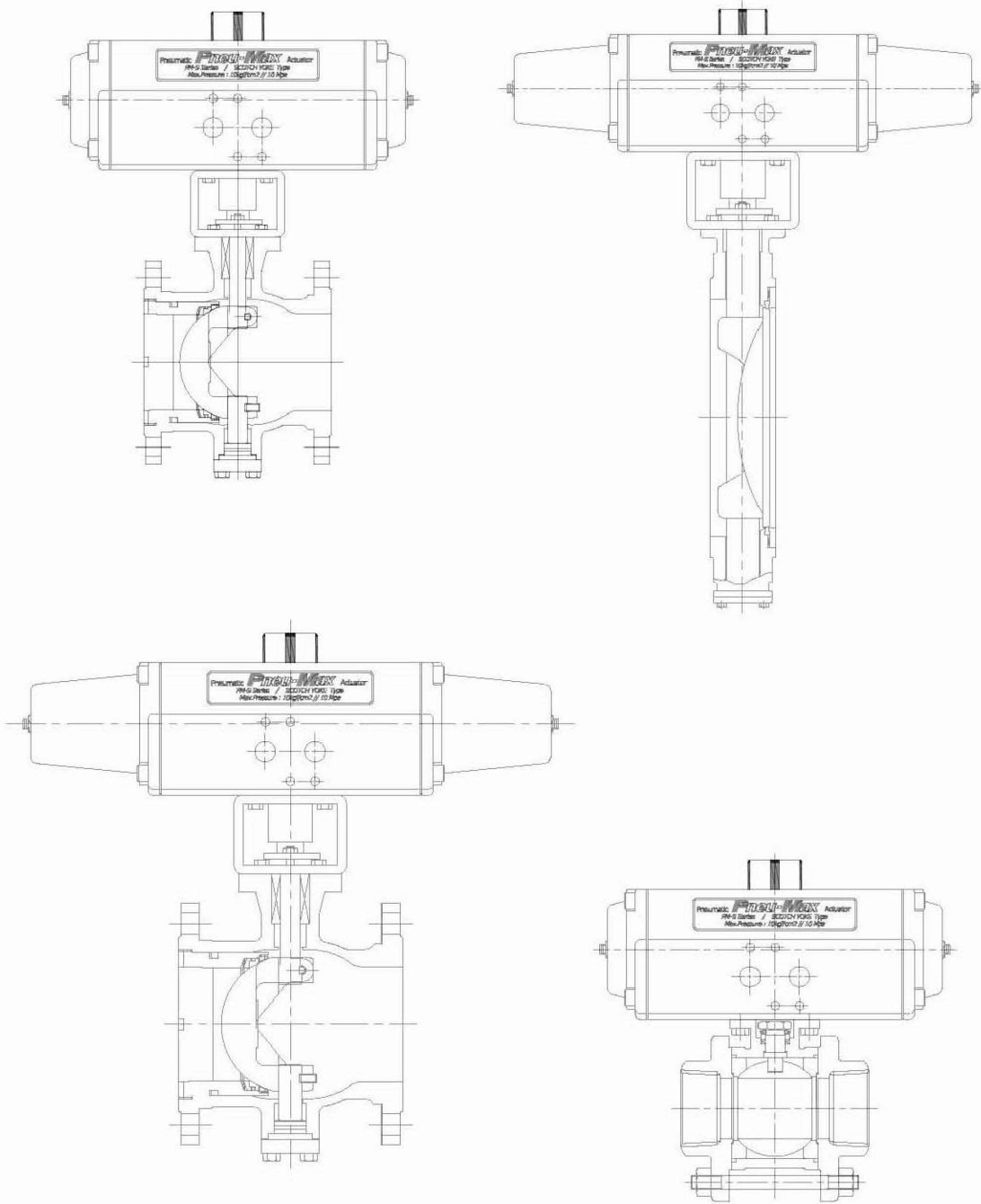
S	Side Hand Wheel
D	Declutchable Hand Wheel

ACTING TYPE

S	Spring Return
D	Double Acting



Pneu-Max®



(주) 엠앤씨 엔지니어링



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